

XICHANG, China

WMO: 565710

Lat: 27.9035N

Lon: 102.2678E

Elev: 5246

StdP: 12.12

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	35.5	37.9	13.8	13.5	64.9	17.3	16.0	61.7	15.0	56.6	12.9	58.6	1.5	0	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	90.7	64.2	88.6	64.4	86.5	64.5	70.8	81.3	69.9	80.1	69.2	78.9	6.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	126.9	73.3	67.4	122.8	73.1	66.6	119.3	72.9	38.4	81.1	37.6	80.2	36.9	79.0	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.0	9.9	DB	31.7	94.6	2.2	3.0	30.1	96.8	28.8	98.5	27.5	100.2	25.9	102.4	(4)
(5)			WB	28.2	72.7	2.4	1.7	26.5	73.9	25.1	74.9	23.8	75.8	22.0	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	51.8	56.3	62.6	68.5	71.9	72.3	74.2	74.3	69.9	64.8	58.1	51.9	(6)
(7)		DBStd	9.91	5.43	7.44	6.62	6.83	7.05	5.66	4.30	4.26	5.64	5.30	4.86	4.52	(7)
(8)		HDD50	113	41	29	5	0	0	0	0	0	0	0	6	32	(8)
(9)		HDD65	1559	409	250	121	45	23	7	1	1	19	69	208	406	(9)
(10)		CDD50	5504	98	206	396	554	679	670	752	752	598	458	250	91	(10)
(11)		CDD65	1474	0	7	47	148	237	227	288	288	167	62	3	0	(11)
(12)		CDH74	10407	1	69	487	1429	2099	1570	1753	1776	912	292	19	0	(12)
(13)		CDH80	3202	0	2	64	456	851	528	491	546	237	27	0	0	(13)
(14)	Wind	WSAvg	3.2	3.4	4.3	4.7	4.4	3.8	2.6	2.4	2.5	2.4	2.4	2.6	2.6	(14)
(15)	Precipitation	PrecAvg	38.2	0.2	0.2	0.4	1.2	3.5	8.2	8.5	7.0	6.3	2.9	0.7	0.2	(15)
(16)		PrecMax	58.0	0.9	1.1	1.8	3.5	8.4	14.9	21.2	16.2	11.1	7.5	3.2	2.0	(16)
(17)		PrecMin	21.9	0.0	0.0	0.0	0.1	0.5	4.3	3.3	2.0	1.5	0.4	0.0	0.0	(17)
(18)		PrecStd	6.4	0.2	0.2	0.4	0.8	1.9	2.5	3.7	3.1	2.2	1.6	0.7	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.4	79.3	84.6	90.6	94.6	93.5	90.2	90.8	89.2	83.0	76.4	70.8	(19)
MCWB			48.8	50.9	54.5	58.6	62.3	66.7	69.1	67.4	67.3	63.8	56.6	49.9	(20)	
2%		DB	69.2	76.1	81.7	87.7	91.3	88.9	87.6	88.1	85.7	80.1	73.6	67.6	(21)	
		MCWB	47.5	49.8	53.5	57.5	61.8	66.8	68.8	67.8	66.9	62.6	55.3	49.5	(22)	
5%		DB	66.6	73.3	79.2	85.3	88.4	85.9	85.2	85.8	82.6	77.4	71.3	65.3	(23)	
		MCWB	46.6	49.0	52.7	57.0	61.7	66.8	68.6	67.7	65.8	61.3	54.6	48.9	(24)	
10%		DB	64.1	70.6	76.6	82.4	85.0	82.6	82.7	83.0	79.6	74.6	68.5	62.7	(25)	
		MCWB	46.1	48.1	51.8	56.5	61.3	66.4	68.4	67.4	65.3	60.6	53.8	48.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	53.6	57.4	62.9	67.6	71.1	72.0	72.0	69.7	66.1	60.3	53.1	(27)
MCD B			66.7	74.3	77.2	82.9	84.2	82.0	82.4	83.9	80.7	76.4	70.6	66.4	(28)	
2%		WB	49.3	51.4	55.6	60.7	66.1	70.0	70.9	70.5	68.3	64.5	58.1	51.1	(29)	
		MCD B	65.3	71.4	76.1	79.6	81.2	80.5	81.0	81.4	79.6	76.2	68.9	64.4	(30)	
5%		WB	48.0	50.1	54.3	59.1	64.8	69.1	70.1	69.5	67.3	63.3	56.3	49.7	(31)	
		MCD B	64.0	70.4	74.8	77.6	79.2	79.0	79.8	79.7	78.0	74.0	67.6	62.8	(32)	
10%		WB	46.6	48.9	53.0	57.9	63.5	68.2	69.3	68.7	66.5	62.0	54.9	48.5	(33)	
		MCD B	62.1	68.3	73.2	76.7	77.8	77.5	78.6	78.3	76.3	71.5	65.9	60.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.1	21.9	22.9	22.0	18.8	14.7	14.2	15.3	14.0	14.5	17.7	18.7	(35)
MCDBR			24.4	25.9	26.6	25.9	24.5	20.7	18.7	19.9	19.3	19.6	22.3	23.4	(36)	
MCWBR			10.3	10.4	9.7	7.7	6.6	5.7	5.4	5.8	5.7	6.2	8.3	9.9	(37)	
5% WB		MCDBR	24.0	25.1	25.2	24.0	20.1	16.1	15.4	16.0	16.2	16.7	19.3	21.9	(38)	
		MCWBR	11.3	10.7	9.7	7.9	6.7	5.3	5.4	5.7	5.0	5.9	7.8	10.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.349	0.462	0.488	0.472	0.466	0.474	0.489	0.471	0.435	0.374	0.345	(40)	
taud		2.369	2.339	1.993	1.984	2.116	2.206	2.181	2.116	2.141	2.182	2.288	2.343	(41)		
Ebn at Noon		288	292	263	260	263	263	261	256	257	259	270	277	(42)		
Edh at Noon		33	37	56	58	51	46	47	50	47	43	36	33	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1188	1465	1624	1716	1648	1343	1372	1337	1129	1058	1121	1048	(44)	
(45)		RadStd	118	161	144	186	140	185	116	169	120	123	142	109	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.74	N/A	-2.74	+1.98	-0.36	N/A	N/A	-166	+342	+205
(47) Regional (4 neighbors)		+0.85	N/A	N/A	+1.69	N/A	N/A	N/A	-166	+342	+205

Nomenclature: See separate page